

BEZHETSK, Russian Federation

WMO: 272170

Lat: **57.800N**Lon: **36.700E**Elev: **147**StdP: **99.57**Time Zone: **3.00 (E03)**Period: **82-93**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.3	-25.3	-32.0	0.2	-29.3	-27.8	0.3	-25.2	10.5	-9.1	9.4	-5.9	2.1	30	0.613

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.6	26.4	19.8	24.7	18.6	23.1	17.5	20.7	25.0	19.4	23.4	18.3	22.1	3.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.1	23.4	17.9	13.0	21.8	16.7	12.1	20.2	60.6	25.3	55.9	23.6	52.3	22.2	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.6	7.6	6.7	-31.2	28.6	4.9	1.3	-34.7	29.6	-37.6	30.3	-40.3	31.0	-43.8	32.0		
			-30.0	22.6	6.1	1.4	-34.3	23.5	-37.9	24.3	-41.3	25.1	-45.7	26.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.8	-9.9	-9.2	-3.0	4.5	11.8	14.9	16.9	15.1	10.1	4.1	-2.9	-7.4
	DBStd	10.95	9.01	7.25	5.26	4.36	4.32	3.79	3.02	3.56	3.50	4.13	5.58	6.61
	HDD10.0	2929	618	539	404	172	32	7	1	4	39	187	387	541
	HDD18.3	5342	876	772	662	414	204	114	64	111	247	442	637	799
	CDD10.0	673	0	0	0	8	89	153	215	161	43	3	0	0
	CDD18.3	43	0	0	0	0	2	11	20	9	1	0	0	0
	CDH23.3	293	0	0	0	0	27	66	131	64	4	0	0	0
	CDH26.7	29	0	0	0	0	1	7	16	6	1	0	0	0
(14) Wind	WSAvg	3.2	3.7	3.6	3.4	3.3	2.9	2.8	2.4	2.4	2.9	3.1	3.6	3.7
(15) Precipitation	PrecAvg	541	25	20	22	34	52	66	78	67	53	46	41	36
	PrecMax	718	55	44	54	102	95	170	189	128	127	86	80	86
	PrecMin	349	2	1	4	1	11	10	24	8	8	0	15	16
	PrecStd	81	13	10	11	18	24	34	37	34	25	21	18	16
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.6	4.9	9.1	19.8	26.1	27.4	28.4	27.3	22.7	15.6	7.4	3.5
		MCWB	1.7	3.5	6.6	13.1	17.9	20.1	21.0	20.2	16.4	12.9	5.7	2.6
	2%	DB	1.9	3.3	5.9	17.0	23.7	25.2	26.8	25.1	19.7	12.7	6.0	2.5
		MCWB	1.1	2.3	3.2	10.6	16.2	19.5	20.3	18.8	15.1	10.7	4.9	1.7
	5%	DB	1.2	1.9	4.4	14.3	21.6	23.4	24.9	23.3	17.8	11.1	4.7	1.7
		MCWB	0.4	1.2	2.3	9.3	14.9	17.9	19.3	17.8	13.9	9.4	3.9	1.1
	10%	DB	0.4	0.6	2.8	11.8	19.7	21.8	23.1	21.6	15.8	9.6	3.5	0.7
		MCWB	-0.3	0.0	1.1	8.1	13.8	16.7	17.8	17.0	12.6	8.3	2.8	0.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.2	3.8	6.8	13.4	18.7	22.3	22.6	21.0	17.5	13.8	6.1	2.9
		MCDB	2.5	4.7	9.0	19.6	24.6	27.3	27.0	26.4	20.7	15.0	6.9	3.3
	2%	WB	1.2	2.4	4.0	11.6	17.0	19.9	21.3	19.5	15.7	11.2	4.9	2.0
		MCDB	1.6	3.0	5.3	16.0	21.7	23.8	25.5	23.9	18.9	12.4	5.9	2.3
	5%	WB	0.5	1.2	2.5	10.0	15.7	18.7	19.8	18.3	14.3	9.8	4.0	1.1
		MCDB	1.0	1.8	3.7	13.5	20.4	22.6	23.7	22.2	17.1	10.9	4.7	1.6
	10%	WB	-0.2	0.1	1.5	8.3	14.4	17.4	18.6	17.4	13.0	8.4	2.9	0.2
		MCDB	0.4	0.8	2.6	11.3	18.7	20.7	22.1	20.8	15.5	9.4	3.4	0.8
(35) Mean Daily Temperature Range	5% DB	MDBR	6.1	6.7	6.8	7.9	9.7	8.4	8.6	8.7	7.5	5.7	4.7	5.8
		MCDBR	4.3	4.0	8.1	12.1	12.5	10.8	11.4	11.7	10.4	8.0	4.2	5.0
	5% WB	MCWBR	4.4	3.4	5.7	7.0	6.3	5.6	5.7	6.4	6.2	5.7	3.6	4.8
		MCWBR	4.4	3.4	6.1	11.1	11.3	10.0	10.5	10.6	9.8	6.9	4.1	5.1
(40) Clear-Sky Solar Irradiance	taub	taub	0.278	0.318	0.336	0.387	0.376	0.366	0.402	0.401	0.363	0.340	0.309	0.269
		taud	2.091	2.066	2.121	2.192	2.358	2.401	2.320	2.310	2.395	2.365	2.214	2.084
	Ebn at Noon	Ebn at Noon	547	687	797	817	858	872	827	798	776	682	515	437
		Edh at Noon	54	89	111	123	111	108	115	109	86	67	49	41
(44) All-Sky Solar Radiation	RadAvg	0.46	1.27	2.79	3.97	5.25	5.46	5.32	4.17	2.56	1.13	0.39	0.24	0.06
	RadStd	0.08	0.23	0.33	0.43	0.41	0.45	0.66	0.36	0.33	0.20	0.06	0.06	0.06

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (8 neighbors)	+0.69	N/A	N/A	N/A	N/A	N/A	N/A	-247	N/A	N/A

Nomenclature: See separate